

ability to independently control the sample area and aperture size. The sample size is the area that the scanner encoder reads to create an individual pixel. The aperture is the actual opening that allows light into the optical bench of the scanner. The ability to control the aperture and sample size separately is particularly useful for smoothing the film grain when scanning black-and-white and colour negative originals.

While drum scanners are capable of scanning both reflective and transmissive artwork, a good-quality flatbed scanner can produce excellent scans from reflective artwork. As a result, drum scanners are rarely used to scan prints, now that high quality inexpensive flatbed scanners are readily available. Film, however, is where drum scanners continue to be the tool of choice for high-end applications. Because film can be wet-mounted to the scanner drum and because of the exceptional sensitivity of the PMTs, drum scanners are capable of capturing very subtle details in film originals.

Currently drum scanners are obsolete and only a few companies continue to manufacture them. While prices of both new and used units have come down over the last decade, they still require considerable monetary investment when compared to CCD flatbed and film scanners. However, drum scanners continue to stay in demand due to their capacity to produce scans that are superior in resolution, colour gradation, and value structure. Also, since drum scanners are capable of resolutions up to 12,000 ppi, their use is generally recommended when a scanned image is going to be enlarged.

In most current graphic-arts operations, high quality flatbed scanners have replaced drum scanners, being both less expensive as well as faster. However, drum scanners continue to be used in high-end applications, such as museum-quality archiving of photographs and print production of high-quality books and magazine advertisements. In addition, due to the greater availability of pre-owned units many fine-art photographers are acquiring drum scanners, creating a new niche market for these machines.